

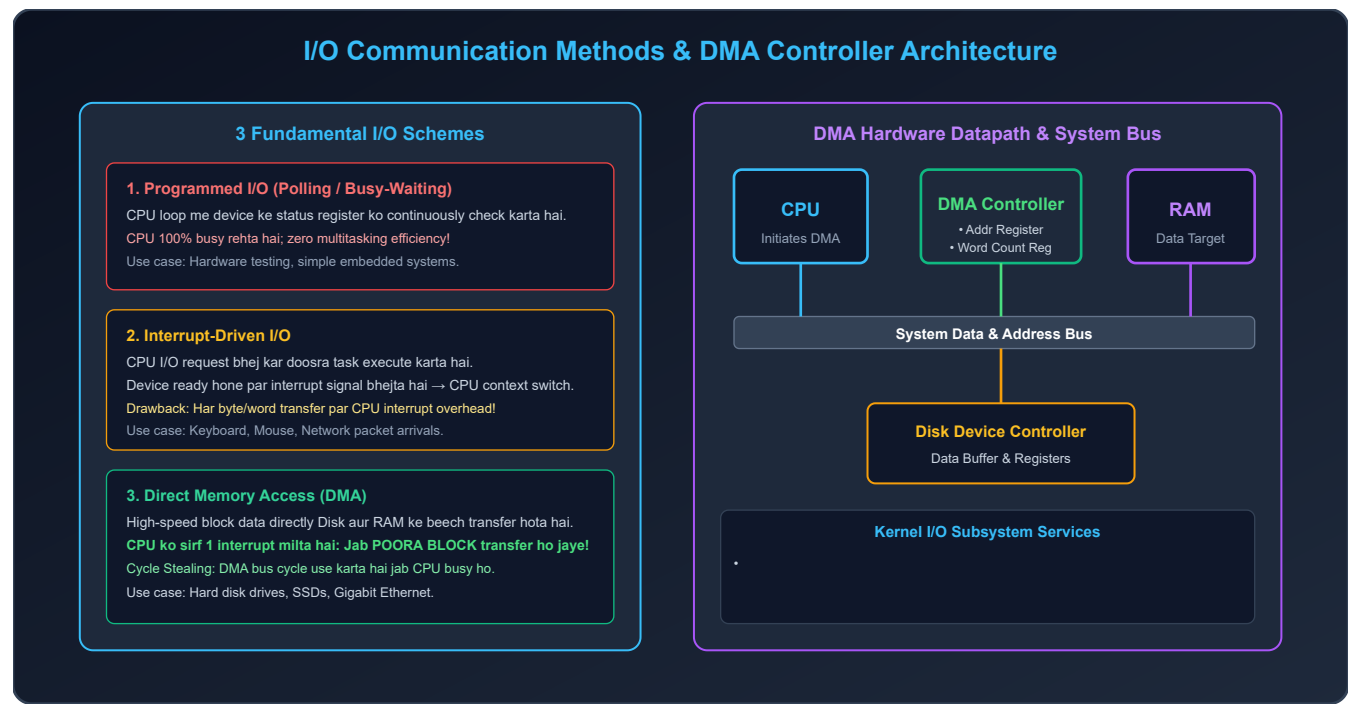
Module 01: I/O Hardware and Kernel Subsystems

1. I/O Transfer Techniques

Method	CPU Involvement	Data Transfer Path	Primary Use Case
Polling (Programmed)	100% Busy Waiting Loop	Device → CPU Register → RAM	Simple dedicated microcontrollers
Interrupt-Driven	Interrupt on every byte/word	Device → CPU Register → RAM	Low-bandwidth devices (Keyboard, Mouse)
DMA (Direct Memory Access)	Single interrupt per BLOCK	Device ↔ RAM (Direct bus)	High-speed bulk storage (HDD, SSD, NIC)

2. DMA Controller Architecture

CPU DMA controller me Base Address aur Word Count load karke release kar deta hai. DMA controller cycle stealing technique use karke high-speed data transfer execute karta hai.



Buffering: Speed mismatch handle karna (Single, Double, Circular) • **Caching:** Fast memory me frequent blocks rakhna • **Spooling:** Multi-user simultaneous print queue handle karna

3. Kernel I/O Subsystem Core Services

- **Buffering:** Speed mismatch aur peak load handle karna (Single, Double, Circular).
- **Caching:** Fast RAM me frequently accessed disk blocks maintain karna.
- **Spooling:** Exclusive devices (Printers) ke multi-user requests ko disk queue me serialize karna.